

MODEL SELECTION & TOKEN ECONOMICS

Haiku Claude 3.5 Haiku · 200K · Lowest cost

Fastest. Best for high-volume, simple work: summaries, classification, extraction, quick Q&A. Use it like an intern — great throughput, not your deepest thinker.

Lite Amazon Nova Lite · 128K · Very Low cost

Cheapest option. Ideal for repetitive batch tasks, basic reformatting, and simple lookups where quality tolerance is high.

Sonnet Claude 4 Sonnet · 200K · Moderate cost ★ [Start here](#)

The default for all knowledge work. Strong reasoning, detailed writing, code review, policy interpretation. Best value across quality, speed, and cost.

Pro Amazon Nova Pro · 128K · Moderate cost

Multimodal strength: text + images + video. Use when your documents contain charts or images that need interpretation.

Opus Claude 4 Opus · 200K · Highest cost

Maximum accuracy. Reserve for regulatory interpretation, multi-step agentic workflows, and decisions where a single error has real cost. Overkill and expensive for simple tasks.

TOKEN CHEAT SHEET

1 token	≈ ¾ of a word
200-word email	≈ 270 tokens
10-page report	≈ 6,700 tokens
RAG injection (5 chunks)	≈ 2,500 tokens
Max output / response	≈ 48,000 words

POWER PROMPTING TECHNIQUES

Cost tip Use Haiku/Lite for bulk; Sonnet for analysis; Opus for critical Chain-of-thought

Append "think step by step" or "show your reasoning before answering" to any analytical prompt. Accuracy on multi-step problems improves significantly — the model surfaces its logic, which you can also verify.

Few-shot examples

Show 1–2 examples of the exact output you want before requesting yours. "Here is a good example of the format I need: [X]. Now produce the same for [Y]." Examples outperform instructions alone.

Negative constraints

State what NOT to do explicitly — models process negatives reliably. "Do NOT include generic advice. Do NOT exceed 250 words. Do NOT use bullet points. Do NOT reference competitors."

Structured output

"Return JSON with keys: risk, likelihood (1–5), impact (1–5), mitigation." Structured output is machine-readable, copy-paste ready, and eliminates interpretation errors.

Iterative refinement

Treat one conversation as one working session. Build on responses: "Strengthen the opening" → "Cut to half the

THE CRAFT PROMPT FRAMEWORK

Structure every prompt with these five elements. A complete CRAFT prompt reduces back-and-forth by 60–80% and consistently produces output you can use immediately.

C Context

Who you are, the purpose, constraints, and background. "I am preparing a board presentation on Q2 credit risk. My audience is non-technical executives. The tone must be calm and factual."

R Role

Assign a specific persona with domain expertise. "You are a senior Treasury analyst at Flagstar with 15 years of ALM experience." Role primes the model's perspective and vocabulary.

A Action

A precise, active verb: analyze / draft / compare / extract / evaluate / rewrite / translate. Avoid: 'help me with' or 'tell me about' — these are vague and produce generic responses.

F Format

Specify structure explicitly: "Return a markdown table with four columns: Risk Factor, Likelihood, Impact (1–5), Mitigation." Or: "Three bullet points, max 20 words each, no jargon."

T Tone & Target

Define audience and reading level. "Plain English for a non-finance VP" produces a completely different response than "technical language for the ALCO committee."

ANTI-PATTERNS — WHAT NOT TO DO

✗ Vague ask

"Help me with this" → Be explicit about task, format, and audience

✗ Using Opus for simple work

Reserve Opus for high-stakes analysis. Haiku costs ~50× less

✗ Restarting instead of refining

"Make it more concise" beats starting a new chat from scratch

✗ No context provided

The model cannot tailor output if it doesn't know who will read it

✗ One giant prompt

Break complex work into steps — each step gets a focused, quality response

✗ Accepting the first draft

AI first drafts are starting points. Always refine at least once

✗ Uploading without a question

"Here is a file" alone gives the model nothing to retrieve against

✗ Ignoring stale context

Long chats drift. Start fresh and summarize key context from the prior one

KEYBOARD SHORTCUTS & CHAT COMMANDS

/command Invoke a saved prompt template with an interactive form

HOW RAG WORKS — UNDER THE HOOD

When you upload a document, StarIQ doesn't "read" it the way a human does. Here is the exact pipeline from upload to cited answer:

- 1 Extract**
Apache Tika 3.2 extracts text from 1,000+ formats: PDF, Word, Excel (all sheets), PowerPoint, HTML, email (.eml/.msg), ZIP archives, images (OCR). Formulas extract as computed values. Charts and embedded images are NOT extracted.
- 2 Chunk**
Text is split into ~1,500-character segments with configurable overlap. Overlap preserves meaning across section boundaries — critical for legal contracts, policy documents, and any content where context flows across paragraphs.
- 3 Embed**
Each chunk is converted to a vector — a mathematical fingerprint of its semantic meaning — using Cohere Embed v4. Similar meaning = similar vector, regardless of exact wording.
- 4 Store**
Vectors are stored in PGVector (PostgreSQL) inside Flagstar's private AWS. Fully encrypted at rest (AWS KMS). Never sent externally. Full audit trail for all access.

BUILDING CUSTOM AI AGENTS (WORKSPACE → MODELS)

Your question is also embedded. The system finds the top 5 chunks whose vectors are most semantically similar to your query — not keyword matching, but meaning matching. The result: a purpose-built team member with a defined job, scope, and personality.

- 6 Answer**
Simple build — Treasury Policy Assistant.
The 5 chunks (~2,500 tokens) are injected into the model's context window alongside your question. The model generates a response grounded in those chunks, departmental policies, showing the exact source document and section.
Base Model
Some models are more cost-effective for daily departmental use.
System Prompt
"You are a Treasury policy expert at Flagstar. Answer only using attached knowledge bases. Always cite exact sources. **Focused Retrieval (default)** so explicitly. Today is {{CURRENT_DATE}}, user is {{USER_NAME}}." Sends only the top 5 most relevant chunks per query (~2,500 tokens). Best for large collections. Efficient, fast, and cost-effective.
Knowledge bases
Attach: 'FTP Procedures', 'ALM Policy', 'ALCO Minutes'.
Full Context
Leave others unattached to prevent off-topic retrievals.
Capabilities
Injects the complete file every message — bypasses RAG engine. Citations (required for trust). Disable all image generation. Guides interpret templates. Impractical and expensive for anything over ~3,000 words.
Prompt starter chips
Add clickable suggestions: 'Summarize FTP methodology', 'What is our policy on duration gap limits?', 'Explain ALCO'.

FILE LIMITS & FILE TYPES
Max file size 50 MB

KNOWLEDGE BASE BEST PRACTICES

- Organize by domain**
Separate KBs by topic: "Compliance Policies", "ALM Procedures", "ALCO Minutes". Focused scope = higher retrieval precision and less risk of pulling from irrelevant documents.
- Write precise KB descriptions**
The description trains AI discovery. Write it like a search query: "Flagstar FTP methodology, ALM policies, interest rate risk procedures updated 2024–2025." Vague names produce vague retrievals.
- Keep documents current**
Outdated documents produce outdated answers. Assign a KB owner per department. Run a quarterly review: remove stale files, add updated versions, re-test retrieval quality.
- Test before deploying to a team**
Ask 5–10 representative questions and verify that citations pull from the correct sections. If retrieval is weak, improve document structure — use clear headings and avoid large table-only files.
- Scope custom agents tightly**
When building a custom model, attach only the KBs it actually needs. An unscoped agent can retrieve from ANY KB the user has access to, which increases the risk of hallucination from irrelevant sources.

PROMPT LIBRARIES & SLASH COMMANDS

extracted. Excel formulas show only computed values. Prompts are reusable templates with interactive form variables. Scanned PDFs require Tesseract OCR — currently disabled. Users type / in chat to browse and invoke them. When triggered, a form appears — users fill in the fields and receive structured, consistent output every time.

- Private by default**
Every KB starts as private. Explicitly grant access to RBAC groups or individuals. For KBs containing confidential content, use private visibility and audit access grants quarterly.
Variable types:
`{{field_name}}` Plain text input
`{{name | textarea}}` Multi-line text area
`{{select:options=[...]}}` Dropdown ("High", "Med", "Low")
`{{due | date:required}}` Date picker — required field
`{{urgent | checkbox}}` Boolean toggle
`{{amount | number:min=0}}` Numeric input with min/max
`{{CURRENT_DATE}}` Auto-injected: today's date
`{{USER_NAME}}` Auto-injected: the requestor's name
`{{CLIPBOARD}}` Auto-injected: user's clipboard content

MULTI-MODEL WORKFLOW

Switch models mid-conversation with @ModelName to run different phases at the optimal cost-quality point:

@Haiku

Extract all dates, amounts, and party names from the contract (fast, cheap)

@Sonnet

Analyze extracted terms for risk, ambiguity, and missing

RBAC & SHARING STRATEGY

All Workspace resources share an additive permission model. Permissions NEVER deny — if ANY group a user belongs to grants a permission, they have it. Design accordingly: minimize Global Defaults, grant specifics via groups.

Permission Groups (for feature rights)

Grant specific capabilities: image generation, code interpreter, web search. Set 'Who can share' = Nobody so they stay invisible in sharing menus. Examples: 'Power Users', 'Developers', 'Data Scientists'.

Sharing Groups (for resource access)

Organize users by team or project for sharing models, KBs, and prompts. Set 'Who can share' = Members (private) or Anyone (open). Examples: 'Treasury Team', 'HR Leadership', 'Risk Committee'.

PRINCIPLE OF LEAST PRIVILEGE

- Minimize Global Defaults — grant rights through groups, not globally
- Attach only the KBs an agent needs — unscoped models pull from all accessible KBs

DATA CLASSIFICATION & GOVERNANCE

Before uploading any document to StarIQ, apply Flagstar's designated owners. Enforce Data Governance (DGG) Policy classification

- Public** grants Group memberships auto-sync from Microsoft Entra ID on every login. Producible: Annual reports, press releases, regulatory guidance, published industry research.
- Internal** User Roles: Admin (full) · User (standard) · Pending (new SSO accounts) Generally appropriate. Internal procedures, meeting notes, project plans, team presentations. Standard StarIQ use.

- Confidential** Proceed with caution. Non-public financials, strategy documents. Configure KB access controls before uploading. Consult the AI Team for guidance.

- Restricted / Sensitive** DO NOT upload without explicit written authorization. Customer PII, Social Security numbers, account numbers, authentication credentials, material non-public information (MNPI), or GLBA-protected data.

2026 ROADMAP

- Q1 — Jan-Mar 2026** Core platform, all 5 models via Amazon Bedrock, knowledge bases & RAG, prompt libraries with slash commands, voice dictation & voice mode, full accessibility
- Q2 — Apr-Jun 2026** OneDrive & SharePoint RAG integration (auto permission enforcement) · Enterprise Data Governance — data ownership and user responsibilities · Mobile PWA (offline + push notifications) · AI Web Search via Palo Alto Prisma AI firewalls
- Q3 — Jul-Sep 2026** Enterprise Corporate Information Management — defines data

DOCUMENT GENERATION & IN-CHAT RENDERING

Generate downloadable Office files in chat:

- Word (.docx)** — Reports, summaries, proposals, meeting notes, policy drafts
- PowerPoint (.pptx)** — Presentations with speaker notes, layouts, themes
- Excel (.xlsx)** — Data exports, financial tables, tracking sheets, models

Files available to download for 30 days after generation.

In-chat Artifacts (no download, rendered in panel):

- Mermaid** Flowcharts, org charts, Gantt, sequence diagrams, ERDs, pie charts
- HTML** Interactive dashboards, calculators, styled reports
- SVG** Infographics, logos, technical drawings
- CSV / JSON** Quick data tables and syntax-highlighted structured data
- LaTeX** Mathematical equations and scientific notation

BEST PROMPTS FOR DOCUMENT GENERATION

- Specify sections explicitly: "Include headings: Executive Summary, Key Risks, Recommendations"
- Provide your data directly rather than asking the AI to invent numbers
- Request speaker notes for PowerPoint: "Add 2-sentence speaker notes per slide"

AI USE CASE INTAKE PROCESS

Amount, Owner, Status" Everyday use — summarizing, drafting, analyzing — does NOT require taking documents into parts where you paste code, go beyond standard response

SUBMIT AN INTAKE FORM WHEN:

- Building a custom agent that accesses sensitive or regulated data
- Integrating StarIQ with other Flagstar systems, APIs, or databases
- Deploying AI for any customer-facing process or workflow
- Your initiative involves GLBA, SOX, HMDA, or other regulated data
- Evaluating or procuring any third-party AI service or vendor
- Deploying Python-based Tools — must pass security code review first

WHAT ADMINISTRATORS CAN SEE

- Conversation logs** — All chats — required for regulatory audit
- Model usage** — Token consumption, model selection, cost patterns
- KB access** — Who accessed which knowledge base and when
- User activity** — Login times, feature usage, document uploads
- Tool executions** — All Python tool calls and their outputs
- StarIQ does NOT train AI models on your conversations or uploads. Data is used only to generate responses within your session.
- Outdated AI responses** Upload current documents to a KB and reference it; use web search (Q2)

TRUBLESHOOTING QUICK REFERENCE

- SSO login fails** — Who accessed which knowledge base and when
- Access via myapps.microsoft.com** — check OAuth redirect URIs
- Doc missing from KB** — Check Documents tab for processing status; verify # reference in chat